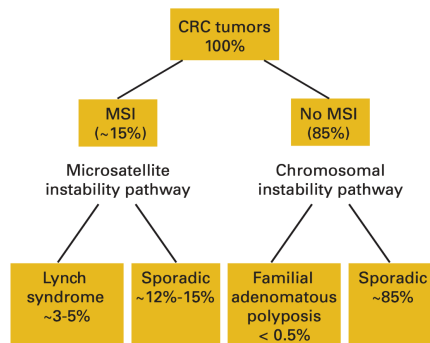


# Genetics in Colorectal Cancers

## Genetics of Colorectal Cancer



(Chapelle and Hampel 2010)

## Lynch Syndrome

- 3% of all colorectal cancers
- Germline mutations of mismatch repair genes
  - *OR* Germline deletions in EPCAM gene
- 80% lifetime risk colorectal cancer
- Early onset ( $\tilde{x}$  45-60yrs)
- Tumors tend to be right-sided

(Eng et al. 2022)

## Mismatch Repair Protein Loss in Lynch Syndrome

Lynch Syndrome can be caused by loss of expression of:

- MLH1
- PMS1
- MSH6
- MSH2
- MSH5

## Mismatch-repair Deficient Colorectal Cancers

Deficiency in mismatch repair proteins → microsatellite instability

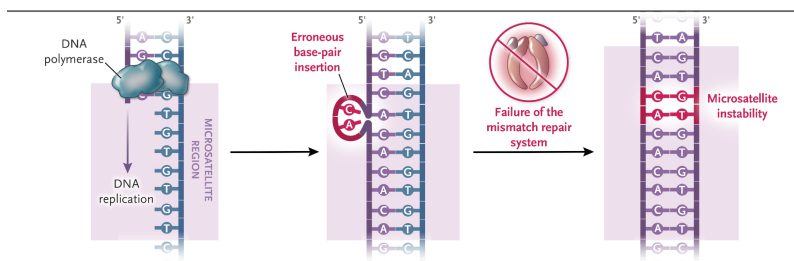
15% of colorectal cancers are MMR-deficient (MSI-High)

- Sporadic - Promoter methylation of MLH1 by BRAF V600E mutation (10%)
- Lynch - Due to mutations in mismatch repair genes (5%)  
90% of Lynch tumors are MMR-deficient

*Reflex testing for BRAF V600E mutation is essential for MMR-deficient tumors*

## Mismatch Repair Proteins and Microsatellite Instability

Deficiency in mismatch repair proteins → microsatellite instability



## Mismatch Repair and MSI Testing

MMR Protein expression can be detected by immunohistochemistry

- Routine testing of tumor for GI cancers at CMC
- Loss of MLH1 leads to loss of PMS2

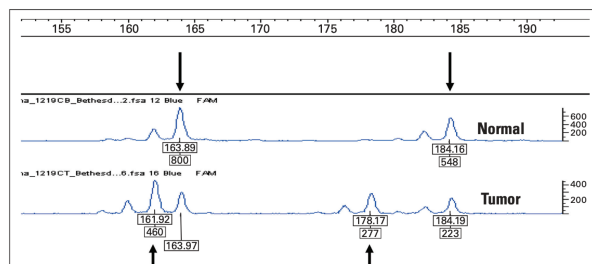
Microsatellite Instability can be detected by pCR

- Tumors are not routinely tested for MSI at CMC

97-99%+ concordance between dMMR and MSI-High status

## Mismatch Repair -> Microsatellite Instability

| Repeat Type     | Sequence                                                 | Name and Location                            |
|-----------------|----------------------------------------------------------|----------------------------------------------|
| Mononucleotide  | AGGTAAAAAAAAAAAAAAAAAAAAAGGGT<br>(A) <sub>24</sub> shown | BAT26, Intron 5 of MSH2 gene<br>Chromosome 2 |
| Dinucleotide    | TGTACACACACACATCGA<br>(CA) <sub>8</sub> shown            | D5S346<br>Chromosome 5                       |
| Tetranucleotide | ATATTCTATCTATCTATCTATCTG<br>(TCTA) <sub>5</sub> shown    | D14S608<br>Intergenic region chromosome 14   |



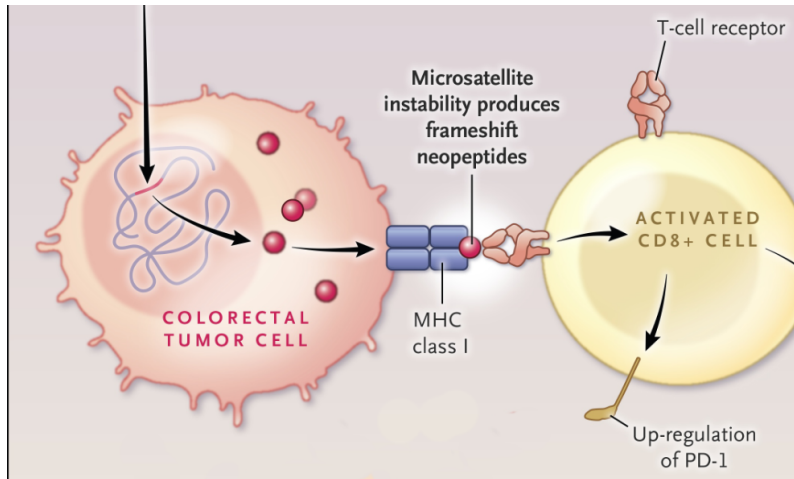
## MMR-deficient Tumor Clinical Features

- Proximal location
- Poor differentiation
- Mucinous histology
- Better prognosis (in early stage)
- Respond well to immunotherapy (ICI)
- Less responsive to 5-FU chemotherapy (in stage II)

(Taieb et al. 2022)

## MMR-deficient Tumors are Immunogenic

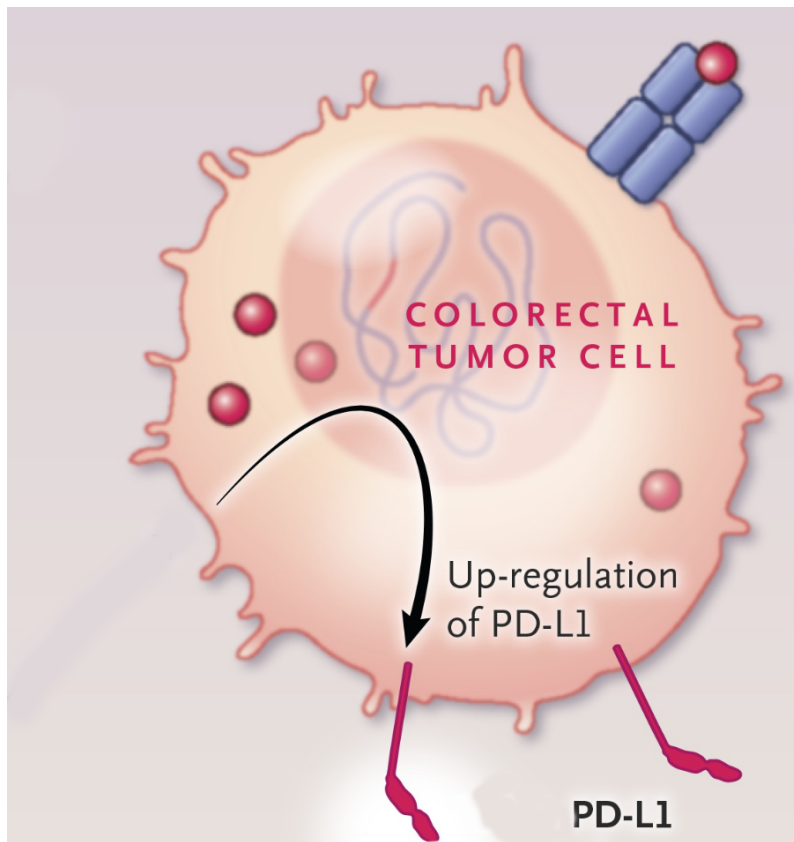
Mutational burden from dMMR causes MSI-High tumors to be immunogenic

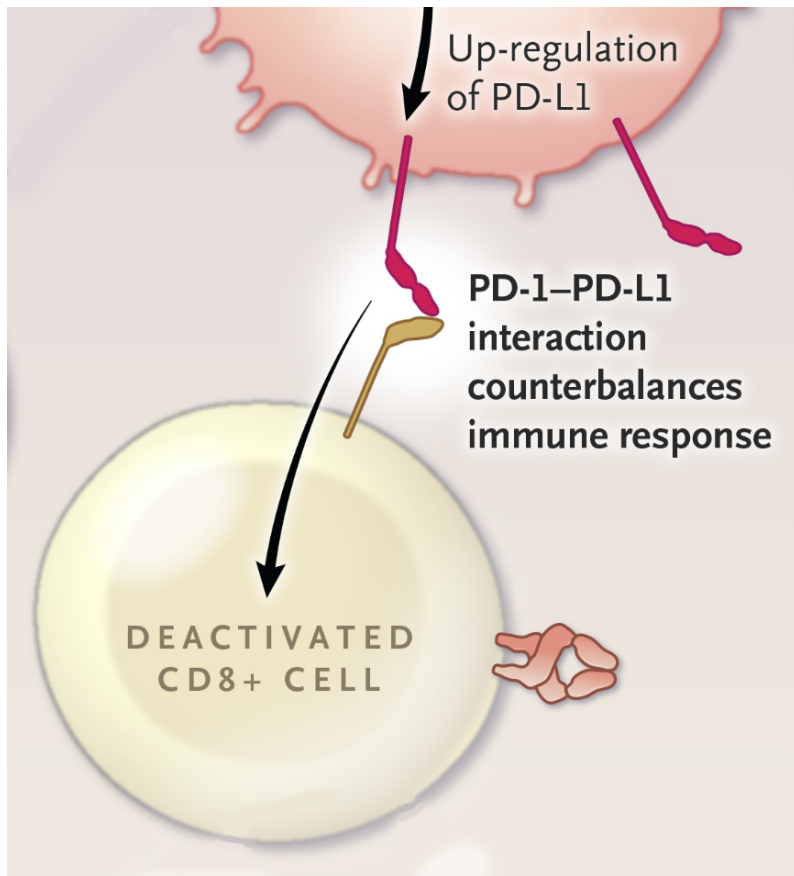


Activated tumor-infiltrating lymphocytes up-regulate PD-1

## Tumor PD-L1 Down-regulates Immune Responses

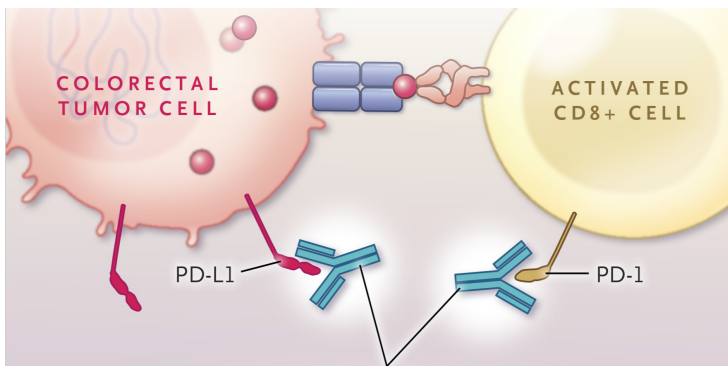
Tumor cells respond to immune activation by  $\uparrow$  PD-L1





↑ PD-L1 downregulates immune response

### Immune Checkpoint Inhibitors



Inhibition of PD-L1  $\leftrightarrow$  PD-1 activates T cells by removing inhibition

### **PD-L1 expression is higher in MMR-deficient CRC**

Immunohistochemistry analysis of 1800 colorectal cancer specimens:

PD-L1 was more often positive in dMMR (18.6%) than in MMR proficient (pMMR) cancers (4.1%;  $p < 0.0001$ )

(Möller et al. 2021)

### **MSI-High Colorectal Cancers**

Clinical Relevance of Microsatellite Instability in Colorectal Cancer. Journal of Clinical Oncology : Official Journal of the American Society of Clinical Oncology. 2010. de la Chapelle A, Hampel H.

### **BRAF V600E Colorectal Cancer Features**

1. Age >70
2. Female
3. Proximal Right sided tumors
4. MLH1 hypermethylation
5. High grade and poor differentiation
6. More peritoneal and nodal metastasis
7. Less lung metastasis

### **Neoadjuvant Rx for MMR-deficient Colon Cancer - NICHE**

115 colon cancer: neoadjuvant nivolumab + ipilimumab  
/ *rightarrow* surgery

109 patients had a pathological response

105 Major pathological response (TRG1 or TRG2)

- 75 pathological complete response

No recurrence of disease at median follow up of 26 months

(Chalabi et al. 2024)

## **KRAS and BRAF**

Don't respond to cetuximab and panitumumab

## **Adjuvant Therapy Stage II MMR-deficient Colon Cancer**

ASCO: No benefit of treatment of stage II dMMR colon cancer with 5-FU alone

Pooled analyses showing no difference in disease-free survival (HR 0.89, 95% CI 0.45-1.74) or overall survival (HR 1.03, 95% CI 0.53-2.02) with adjuvant chemotherapy in this population.

(Ribic et al. 2003)

## **Adjuvant Therapy of Stage III MMR-deficient Colon Cancer**

Oxaliplatin-based chemotherapy (FOLFOX or CapeOX) and atezolizumab (Tecentriq)

ATOMIC trial: atezolizumab + mFOLFOX6 is superior to mFOLFOX6 alone, with a 3-year disease-free survival rate of 86.4% versus 76.6% (hazard ratio 0.50, P .0001)

## **Cancer in Lynch Syndrome**

45 year old with ascending colon cancer with Lynch. Options:

- Neoadjuvant Chemo followed by right hemi
- Right hemi followed by adjuvant 5FU leucovorin
- Total proctocolectomy with end ileostomy
- Total proctocolectomy with ileoanal pouch

- **Total abdominal colectomy with ileorectal anastomosis**

## **FAP screening**

### **Muir-Torre Syndrome - Lynch variant**

- Lifetime cancer risk 80-100%. Multiple primary cancers common (50% have ≥ 2)
- Colorectal: 58% proximal to splenic flexure
- Endometrial (15% of women), ovarian, transitional cell bladder
- Sebaceous: Adenomas (66%) sebaceous epithelioma (33%) sebaceous carcinoma (33%)

### **Muir-Torre Syndrome - Lynch variant**

MSI-positive (65%):

- Similar pathophysiology as Lynch Syndrome
- Strong family history
- Early-onset colorectal cancer (average age 40)
- Prolonged survival
- Multiple cancer sites (visceral and skin)

MSI-negative (35%): Later-onset cancers, less pronounced family history.

- Later-onset cancers
- Less pronounced family history
- MUTYH or MLH1 promoter hypermethylation

Colonoscopy age 10-12 EGD for duodenal polyps at age 20-30  
CT 1-3 years after colectomy and q5 yers in those with family  
hx of desmoids

## **Lynch Non-Colorectal Cancers**

- Endometrial (25-60% penetrance)
- Stomach
- Ovarian
- Urinary tract
- Biliary Tract
- Small bowel
- Male breast cancer (1.2% penetrance vs 0.1%)

(Sinicrope 2018)

## **Lynch Screening**

Colonoscopy q1-2 years starting age 20-25

## **Colon Polyposis: >10 adenomatous polyps**

- Classical FAP
- Attenuated FAP (AFAP)
- MUTYH-associated polyposis (MAP)
- Colonic adenomatous polyposis of unknown etiology (CPUE)

## **Colon Polyposis: >4 hamartomatous polyps**

- Peutz-Jaghers
- Juvenile Polyposis Syndrome
- Cowden Syndrome/PTEN Hamartoma Tumor Syndrome

## **Serrated Colon Polyps**

5 serrated polyps/lesions proximal to the rectum, all being 5 mm in size, with 2 being 10 mm in size OR >20 serrated polyps/lesions of any size distributed throughout the large bowel, with 5 being proximal to the rectum

## FAP

(FAP) is caused by germline mutations in the APC gene and leads to the development of hundreds to thousands of colorectal adenomas, with nearly 100% risk of colorectal cancer if untreated.

## Attenuated FAP

~30 polyps 70% penetrance by age 80 - mean age at dx 50-55

## References

- Chalabi, Myriam, Yara L. Verschoor, Pedro Batista Tan, et al. 2024. “Neoadjuvant Immunotherapy in Locally Advanced Mismatch Repair-Deficient Colon Cancer.” *The New England Journal of Medicine* 390 (21): 1949–58. <https://doi.org/10.1056/NEJMoa2400634>.
- Chapelle, Albert de la, and Heather Hampel. 2010. “Clinical Relevance of Microsatellite Instability in Colorectal Cancer.” *Journal of Clinical Oncology: Official Journal of the American Society of Clinical Oncology* 28 (20): 3380–87. <https://doi.org/10.1200/JCO.2009.27.0652>.
- Eng, Cathy, Alexandre A. Jácume, Rajiv Agarwal, et al. 2022. “A Comprehensive Framework for Early-Onset Colorectal Cancer Research.” *The Lancet. Oncology* 23 (3): e116–28. [https://doi.org/10.1016/S1470-2045\(21\)00588-X](https://doi.org/10.1016/S1470-2045(21)00588-X).
- Möller, Katharina, Niclas C. Blessin, Doris Höflmayer, et al. 2021. “High Density of Cytotoxic T-Lymphocytes Is Linked to Tumoral PD-L1 Expression Regardless of the Mismatch Repair Status in Colorectal Cancer.” *Acta Oncologica (Stockholm, Sweden)* 60 (9): 1210–17. <https://doi.org/10.1080/0284186X.2021.1933585>.
- Ribic, Christine M., Daniel J. Sargent, Malcolm J. Moore, et al. 2003. “Tumor Microsatellite-Instability Status as a Predictor

of Benefit from Fluorouracil-Based Adjuvant Chemotherapy for Colon Cancer.” *The New England Journal of Medicine* 349 (3): 247–57. <https://doi.org/10.1056/NEJMoa022289>.

Sinicrope, Frank A. 2018. “Lynch Syndrome-Associated Colorectal Cancer.” *The New England Journal of Medicine* 379 (8): 764–73. <https://doi.org/10.1056/NEJMcp1714533>.

Taieb, Julien, Magali Svrcek, Romain Cohen, Debora Basile, David Tougeron, and Jean-Marc Phelip. 2022. “Deficient Mismatch Repair/Microsatellite Unstable Colorectal Cancer: Diagnosis, Prognosis and Treatment.” *European Journal of Cancer* (Oxford, England : 1990) 175 (November): 136–57. <https://doi.org/10.1016/j.ejca.2022.07.020>.